

THE ATOMIQ HACKATHON 2026 · FULL TECHNICAL DOSSIER

CertiQ-Chirp

Certified Quantum Campaign Intelligence

Fintech · AI/ML · Qiskit · QAOA · Six-qubit simulation

Six selected customers · Exact conversion probability · 30-metric benchmark

Prepared as a judge-ready, reproducible PDF submission

Document control

Field	Value
Project	CertiQ-Chirp: Certified Quantum Campaign Intelligence
Primary domain	Fintech / targeted term-deposit marketing
Secondary domains	AI/ML, quantum machine learning, constrained optimization
Event	The Atomiq Hackathon | Qiskit Fall Fest 2026 at IEM
Final decision	C0059, C0170, C0256, C0327, C0372, C0374
Submitted Pr(K ≥ 3)	0.420 (Qiskit Aer noiseless finite-shot estimate)
Exact analytical Pr(K ≥ 3)	0.4164131424084104
Core SDKs	Qiskit 2.6.2, Qiskit Aer 0.18.2, scikit-learn, XGBoost, CatBoost, LightGBM
Document version	PDF submission dossier 1.0

Executive summary

CertiQ-Chirp is an auditable hybrid quantum-classical decision pipeline for a bank term-deposit campaign. It predicts calibrated subscription probabilities, selects six customers under budget and capacity limits, and estimates the probability that at least three subscribe with a six-qubit quantum circuit. The design uses Qiskit where it is technically defensible and exact classical references where they make the result verifiable. No quantum advantage is claimed.

PR-AUC (OOF)	Brier (OOF)	Precision@10%	Stage 2 gap	Exact Pr(K≥3)	Net / cost
0.2347	**0.1036**	**0.3000**	**0.0000**	**0.4164**	**4.1625**

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Appendices	Full predictions, candidate data, counts, schemas, and form values

1. Challenge interpretation and source hierarchy

The released technical specification defines one connected pipeline rather than independent quantum demos. The final answers are six customer IDs and Pr(K ≥ 3), where K is the number of selected customers who subscribe. The judges verify Stage 1 against hidden test labels and recompute Stage 2 and Stage 3 exact references from the probabilities submitted.

Priority	Source	Controls
1	Problem statement PDF + workbook	Task, formulas, dataset, constraints
2	Devpost Rules	Dates, eligibility, submission requirements, judging
3	Devpost Overview	Prizes, video, custom form fields
4	Official QFF website	Festival context and programme

Key governing equations

Stage 2 expected net value: $w_i = p_i v_i - c_i$; maximize $\Sigma w_i x_i$ subject to $\Sigma x_i = 6$, $\Sigma c_i x_i \leq 25$, and $\Sigma h_i x_i \leq 22$.

Stage 3 state preparation: $\theta_i = 2 \arcsin \sqrt{p_i}$; success when measured Hamming weight is at least three.

End-to-end certified hybrid pipeline

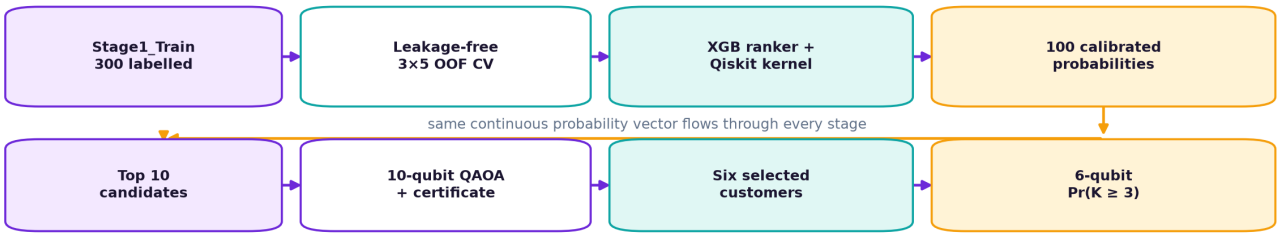


Figure 1: End-to-end architecture. Data, prediction, constrained selection, and quantum simulation form one traceable pipeline.

2. Data inventory and integrity controls

Sheet	Shape	Purpose
README	20 × 2	Field descriptions, constraints, seeds, formulas
Stage1_Train	300 × 17	16 features plus subscription label y
Stage1_Test	100 × 16	Features only; hidden labels held by judges
Stage2_Data	400 × 5	Operational inputs for all 400 customers
Stage2_Constraints	6 × 4	N=10, K=6, budget=25, capacity=22

The 400 customer IDs are unique and span C0001–C0400. Train and test IDs are disjoint. The training set has 35 positive and 265 negative outcomes; by subtraction the hidden test set has 11 positive and 89 negative outcomes. The duration field was removed from the source data. Categorical missingness appears as the literal value unknown, and pdays = −1 is a sentinel.

All preprocessing—scaling, one-hot encoding, PCA, quantum feature angles, model fitting, and probability calibration—was fitted inside repeated stratified training folds. The 100 test customers were predicted once after the pipeline was frozen. The public source dataset was not used to recover hidden test labels.

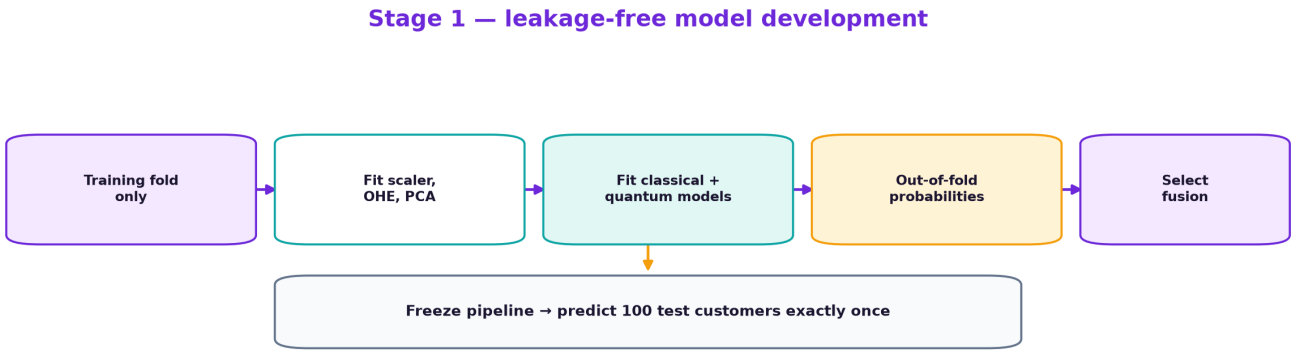


Figure 2: Stage 1 leakage controls. All preprocessing, calibration, and fusion weights are learned from out-of-fold training predictions.

3. System architecture

The architecture separates responsibilities. XGBoost supplies top-decile ranking; a four-qubit Qiskit ZZFeatureMap quantum-kernel SVC supplies a well-calibrated probability prior. Their fusion produces the single probability vector used by every later stage. A warm-start QAOA plus exact certificate produces the six-customer decision, and a six-qubit product-state circuit produces the final campaign probability.

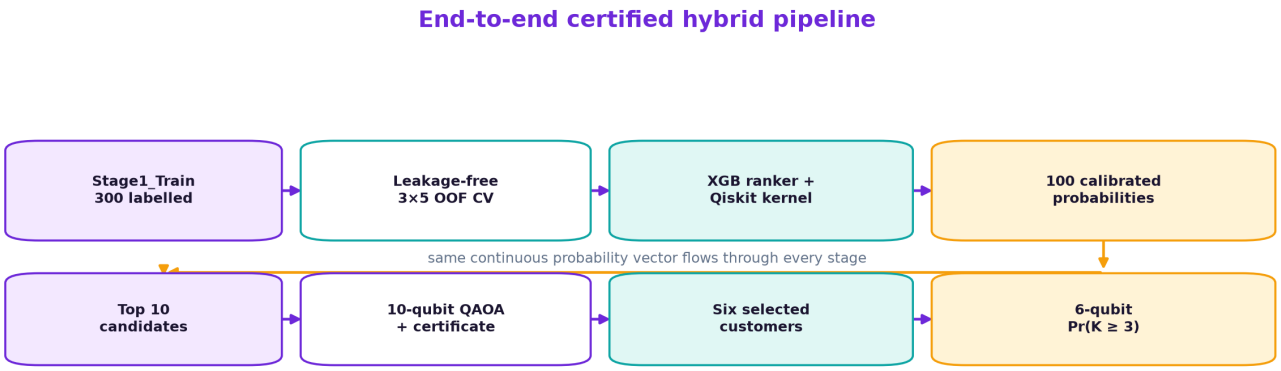


Figure 3: End-to-end architecture. Data, prediction, constrained selection, and quantum simulation form one traceable pipeline.

Quantum / classical boundary

Component	Quantum	Classical
Stage 1	4-qubit ZZFeatureMap kernel, statevector overlaps, SVC	XGBoost, CatBoost, LightGBM, logistic regression, HistGradientBoosting, OOF calibration
Stage 2	10-qubit warm-start QAOA, diagonal cost oracle, measured bitstrings	One-swap repair, 210-subset brute force, exact objective certificate
Stage 3	6 RY qubits, ideal statevector, Aer finite-shot, noisy simulation	64-outcome Poisson-binomial, Monte Carlo, response-matrix mitigation

4. Stage 1 — calibrated hybrid prediction

Repeated 3×5 stratified cross-validation produced out-of-fold predictions for six candidate models and three fusion strategies. The quantum kernel was a weak ranker but a strong calibrator; XGBoost was the strongest top-decile ranker but overconfident. A 60% quantum-kernel / 40% XGBoost fusion achieved the best judge-aligned composite.

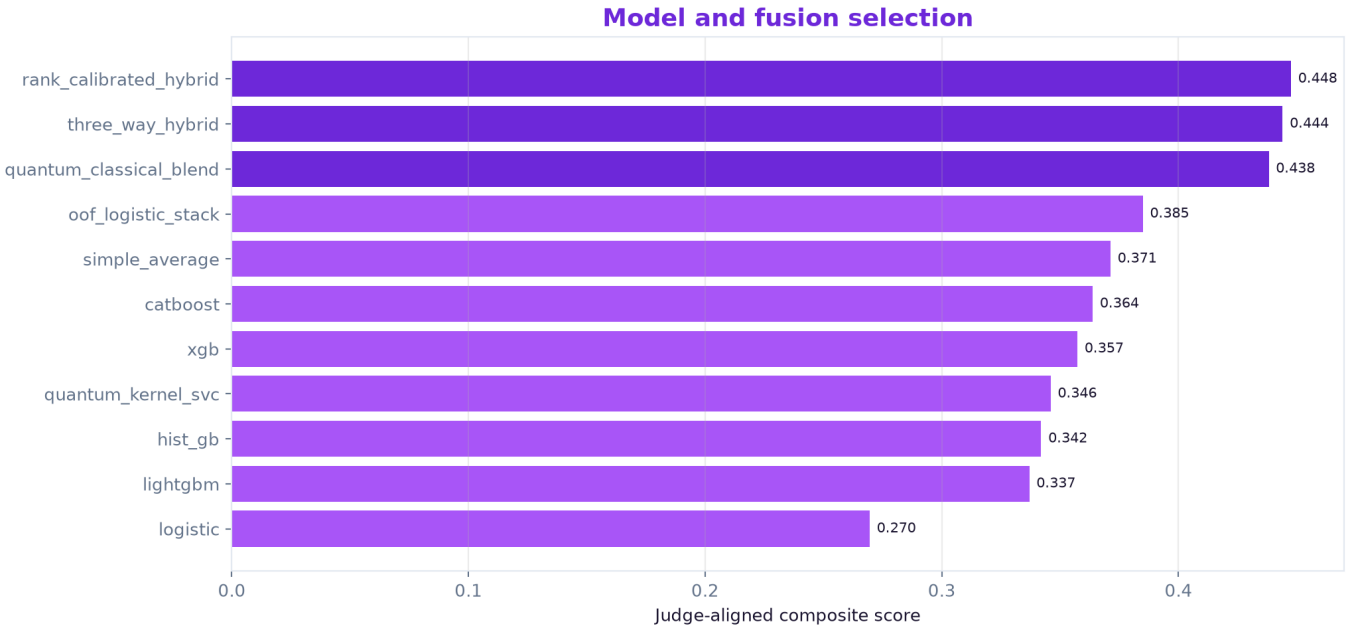


Figure 4: Model selection scoreboard. The 40% XGBoost / 60% quantum-kernel fusion achieved the highest judge-aligned composite.

Metric	Fusion	Best classical	Quantum only
ROC-AUC	0.6069	0.6291	0.4992
PR-AUC	0.2347	0.2243	0.1226
Brier	0.1036	0.1164	0.1039
Log loss	0.3609	0.4023	0.3639
ECE	0.0357	0.0877	0.0342
Calibration slope	0.6124	0.3102	-0.1553
Precision@10%	0.3000	0.2333	0.1333
Recall@10%	0.2571	0.2000	0.1143
Lift@10%	2.5714	2.0000	1.1429
NDCG@10%	0.3628	0.2948	0.0985

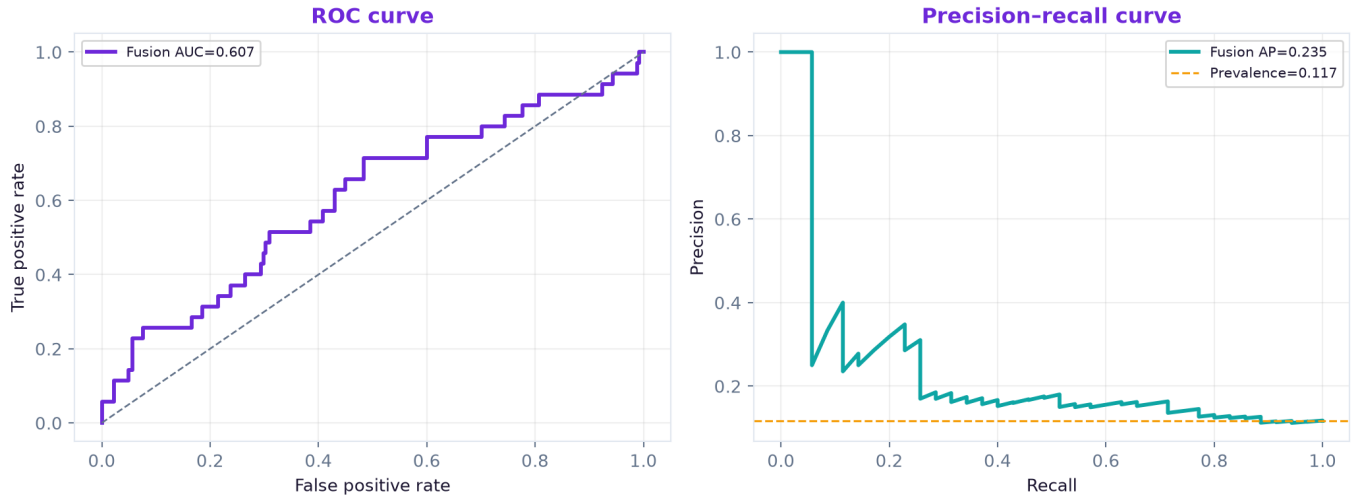


Figure 5: Discrimination curves. Out-of-fold ROC and precision–recall behaviour on the 300 labelled training customers.

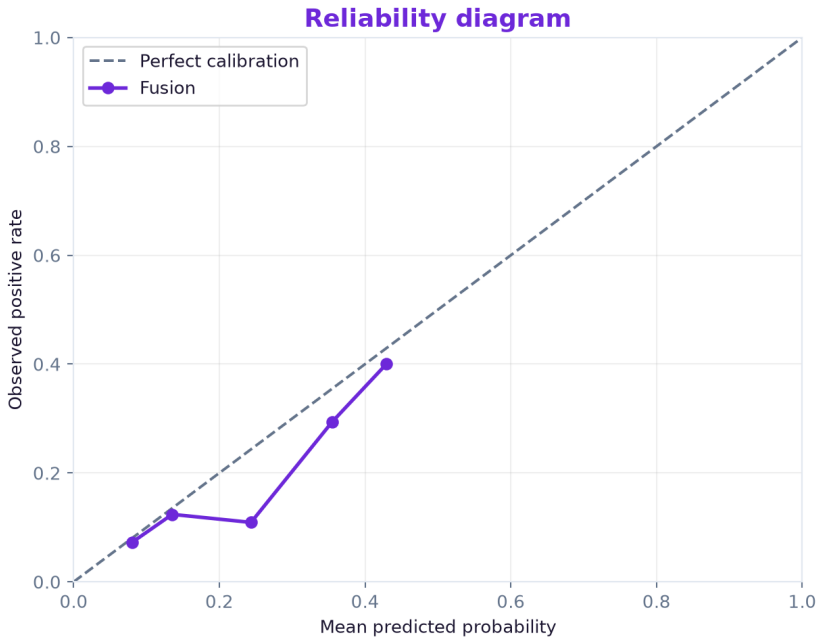


Figure 6: Probability calibration. Reliability curve supports the low Brier score, log loss, and expected calibration error.

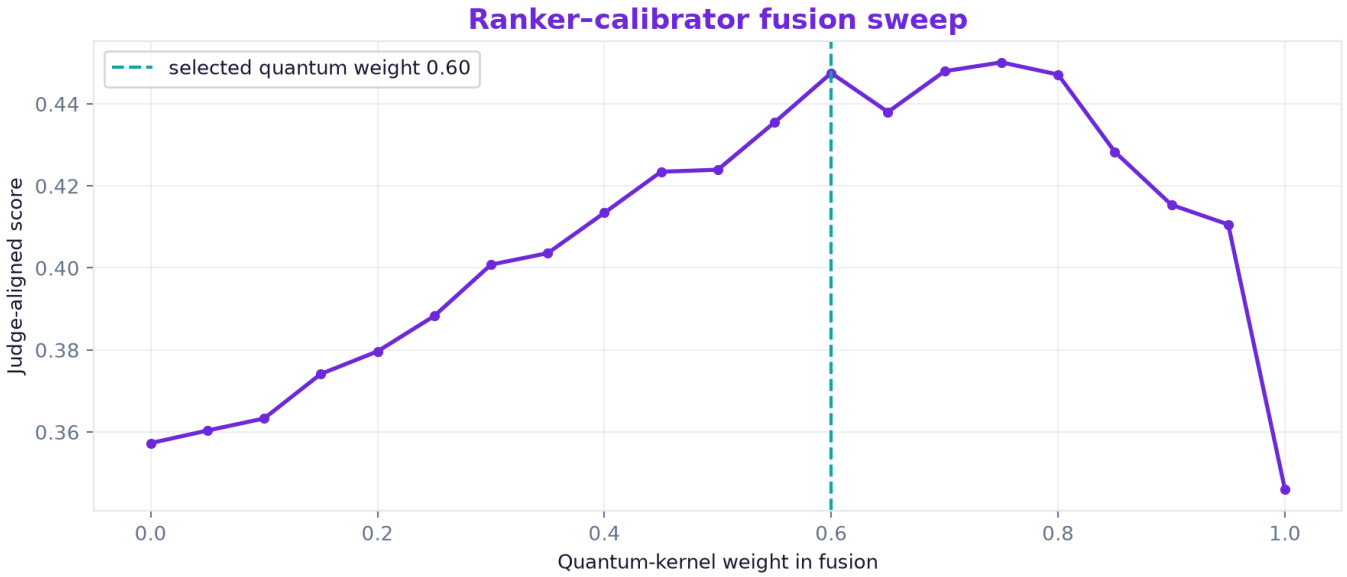


Figure 7: Fusion-weight sensitivity. The selected 60% quantum / 40% XGBoost point balances calibration with top-decile ranking.

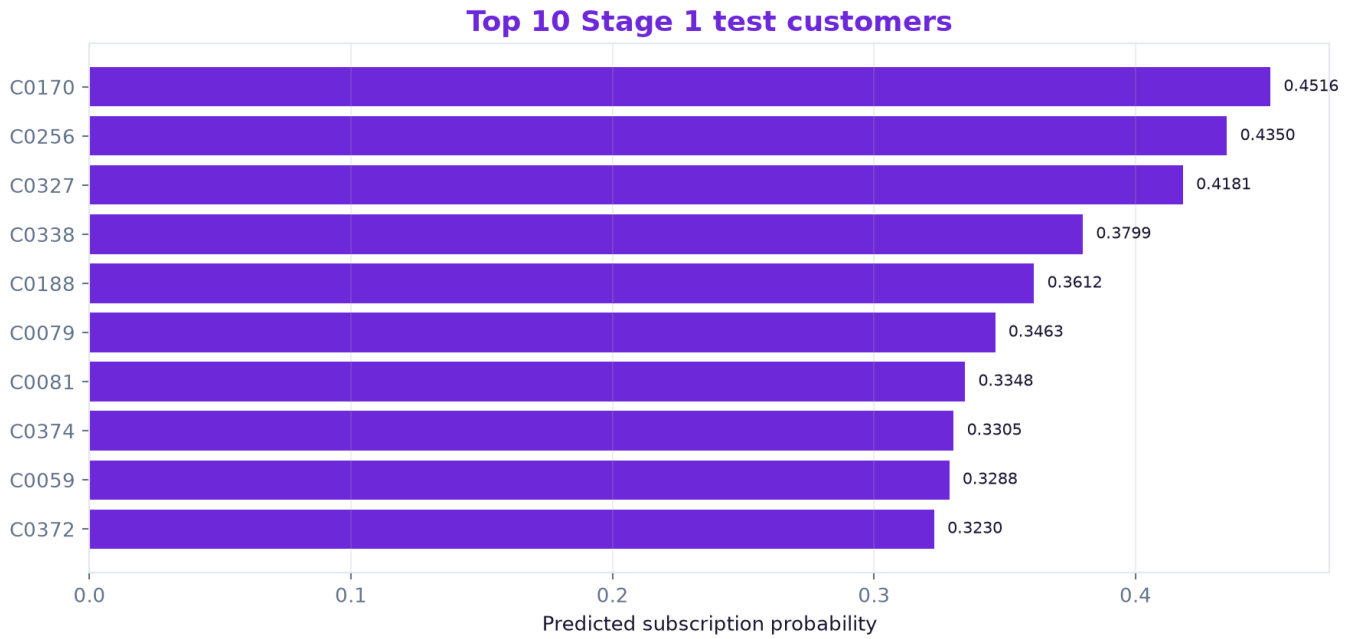


Figure 8: Top-10 candidate probabilities. Continuous probabilities are preserved and ranked with customer_id tie-breaking.

5. Stage 2 — certified quantum-assisted selection

Ten candidates are ranked by Stage 1 probability. For each candidate, expected net value is $p_i v_i - c_i$. The Stage 2 cost Hamiltonian combines the negative objective with exact-six and one-sided resource penalties. The QAOA experiment used ten qubits, two layers, and 8,192 sampled bitstrings. Its raw feasible optimum is reported separately from the certified final result.

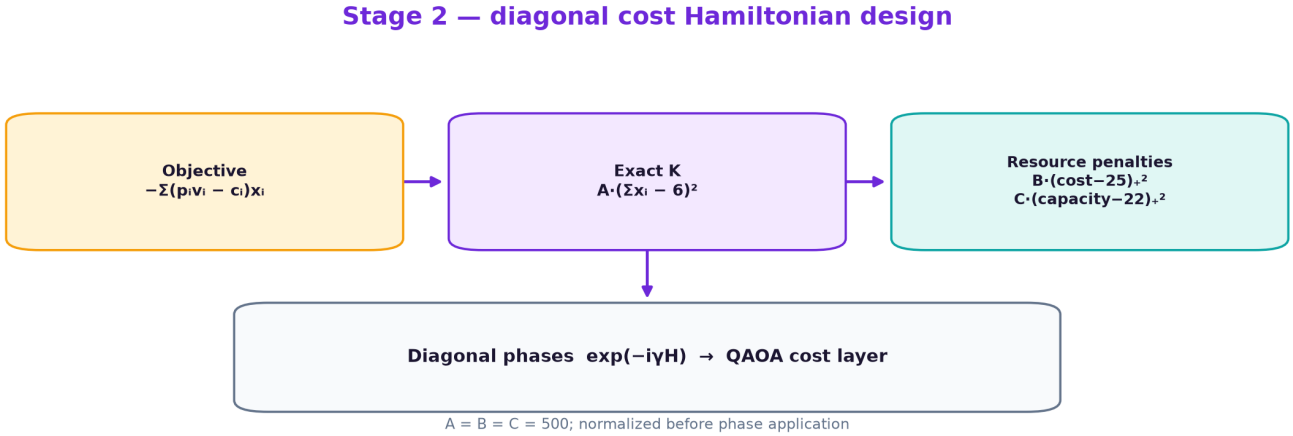


Figure 9: Constraint and penalty design. The exact-six and one-sided resource constraints are encoded in a diagonal QAOA cost oracle.

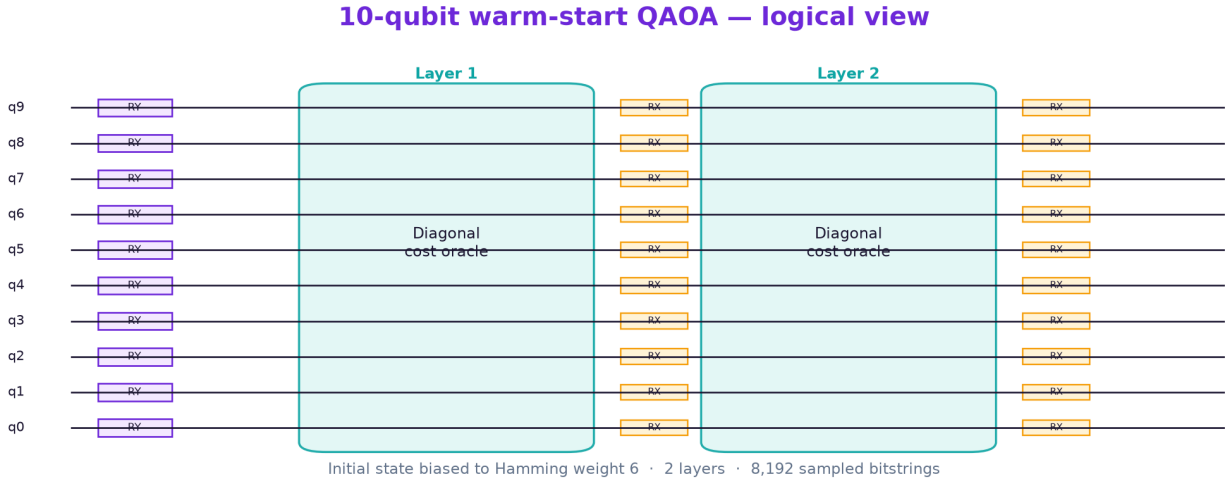


Figure 10: QAOA logical circuit. Warm-start RY initialization, two diagonal cost layers, and RX mixers on ten selection qubits.

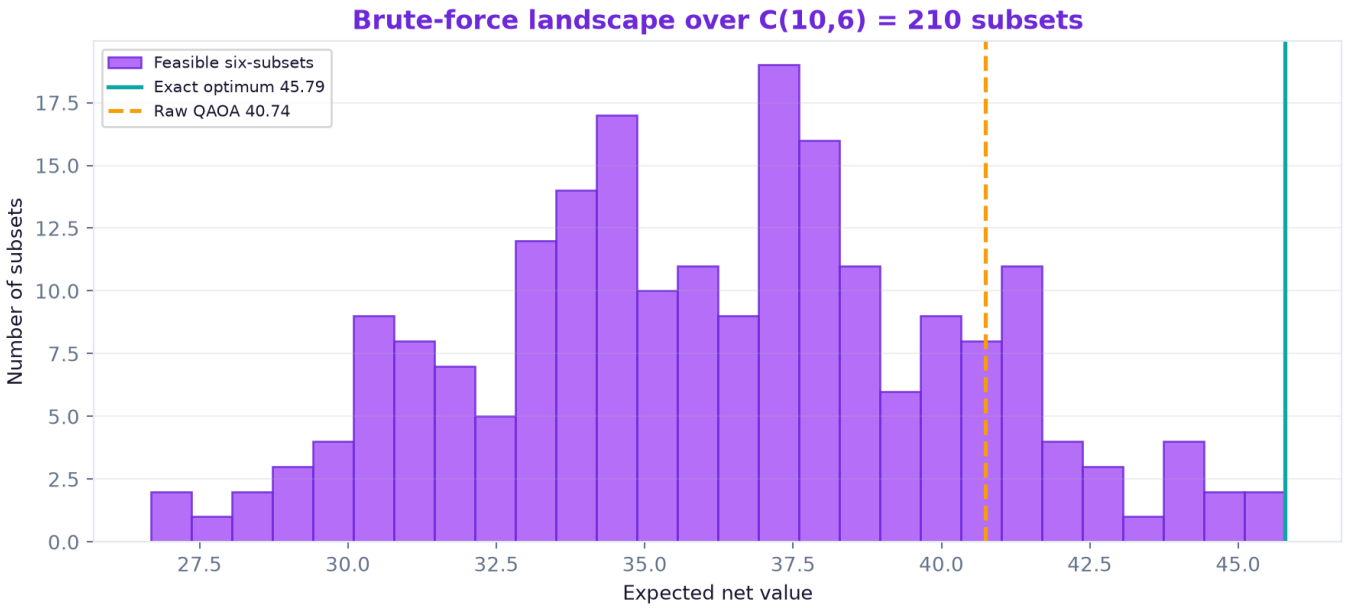


Figure 11: Stage 2 objective landscape. All feasible six-customer subsets are enumerated; raw QAOA and certified optimum are shown.

Quantity	Value
Candidate count	10
Feasible six-subsets	210 / 210
Selected six	C0059, C0170, C0256, C0327, C0372, C0374
Expected net value	45.787439
Budget used	11 / 25
Capacity used	16 / 22
QAOA feasible sample rate	0.681396
Raw QAOA best feasible objective	40.739708
Raw QAOA gap	5.047732
Certified hybrid gap	0.000000

6. Stage 3 — six-qubit conversion simulation

One qubit represents each selected customer. An RY rotation with $\theta_i = 2 \arcsin \sqrt{p_i}$ prepares the subscription probability. The exact reference is the Poisson-binomial tail $\Pr(K \geq 3)$ over 64 independent outcomes. Qiskit Aer supplied noiseless and synthetic-noise measurements; NumPy supplied an independent Monte Carlo benchmark.

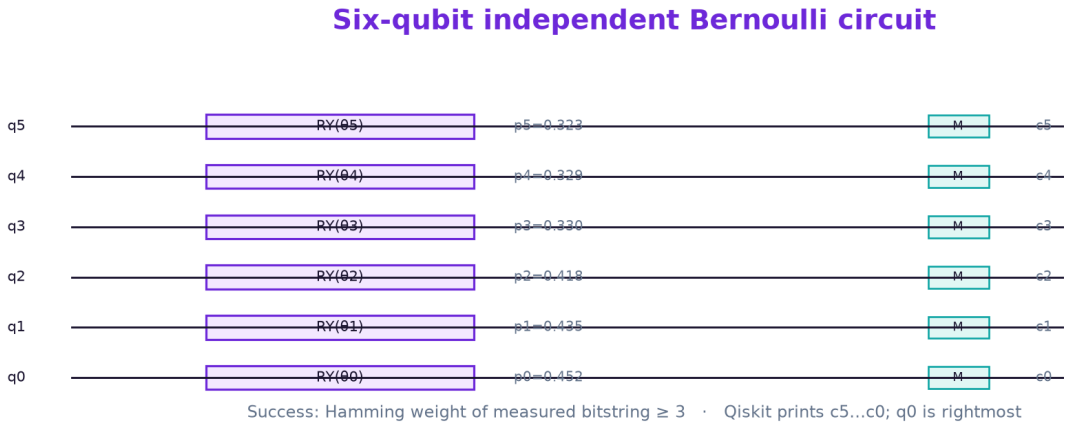


Figure 12: Stage 3 product-state circuit. One RY rotation per customer prepares the exact subscription probability on each qubit.

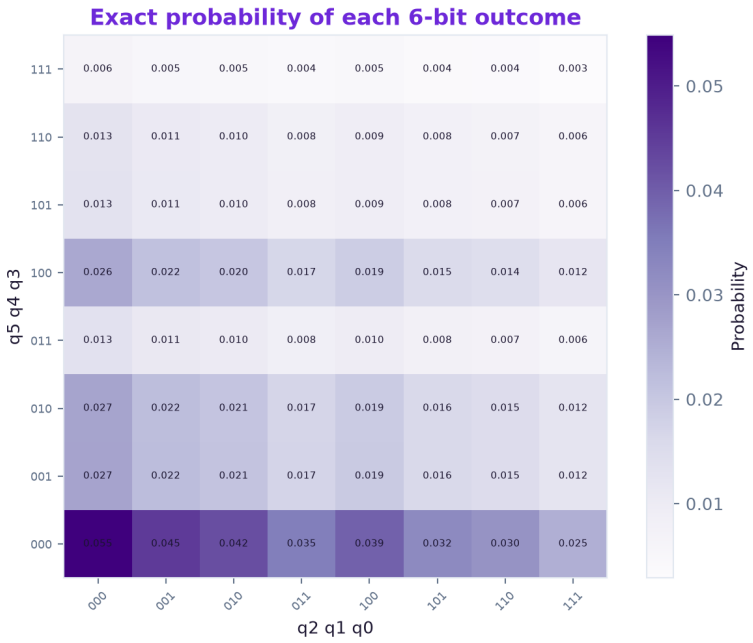


Figure 13: 64-outcome probability map. Independent Bernoulli outcomes; the upper-right region corresponds to three or more successes.

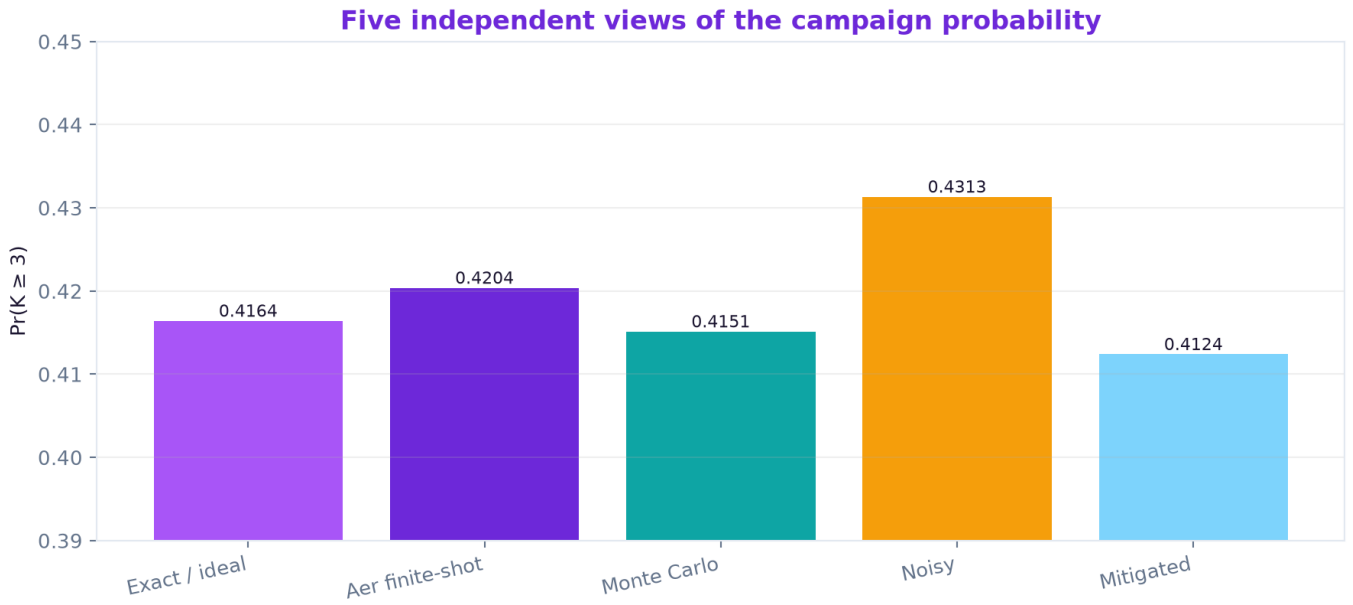


Figure 14: Stage 3 estimate comparison. Exact, ideal statevector, finite-shot, Monte Carlo, noisy, and mitigated results agree within reported errors.

View	Pr(K ≥ 3)	Absolute error
Exact analytical	0.416413	—
Ideal statevector	0.416413	0.000000
Aer noiseless finite-shot	0.420380	0.003966
Monte Carlo	0.415131	0.001283
Noisy Aer	0.431335	0.014922
Mitigated	0.412398	0.004015
64×64 response cross-check	0.411111	0.005302

Noise model: single-qubit depolarizing parameter 0.02 and symmetric readout error 0.03. Primary mitigation inverts the known channel on the six marginals; a 64×64 non-negative least-squares response-matrix unfolding is retained as a cross-check.

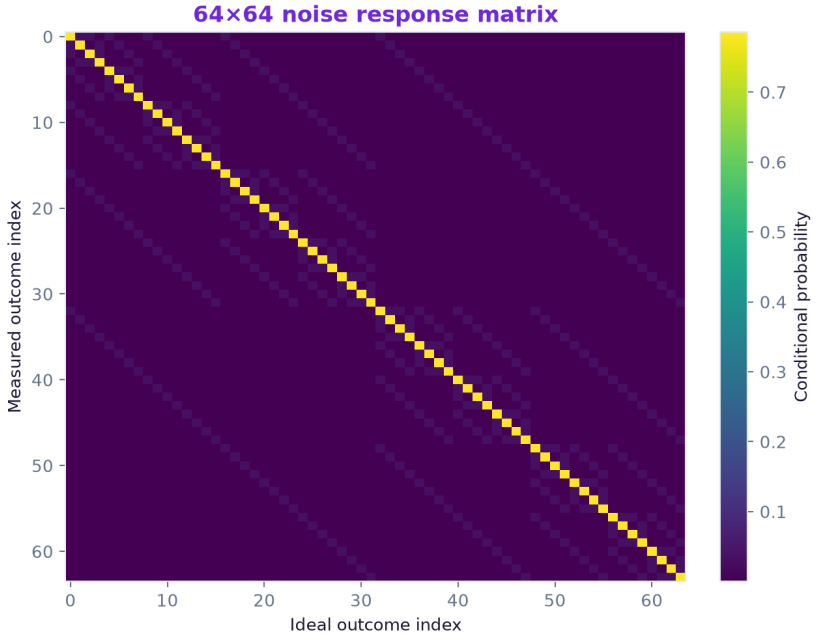


Figure 15: Noise response matrix. Tensor-product response of single-qubit depolarizing noise and symmetric readout error.

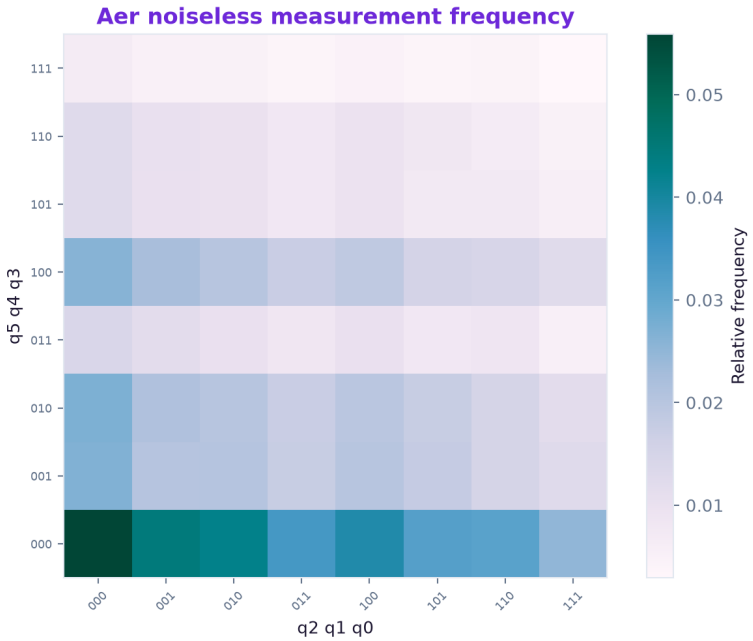


Figure 16: Measurement-count distribution. The 32,768 noiseless shots populate all 64 outcomes according to the product-state distribution.

7. End-to-end integration and error propagation

The same continuous probability vector drives ranking, expected net value, selection, and circuit angles. Customer IDs are traceable from Stage 1 through Stage 3. Calibration uncertainty is summarized with a bootstrap PR-AUC interval and repeated-CV stability, while Stage 2 and Stage 3 errors are quantified against exact references.

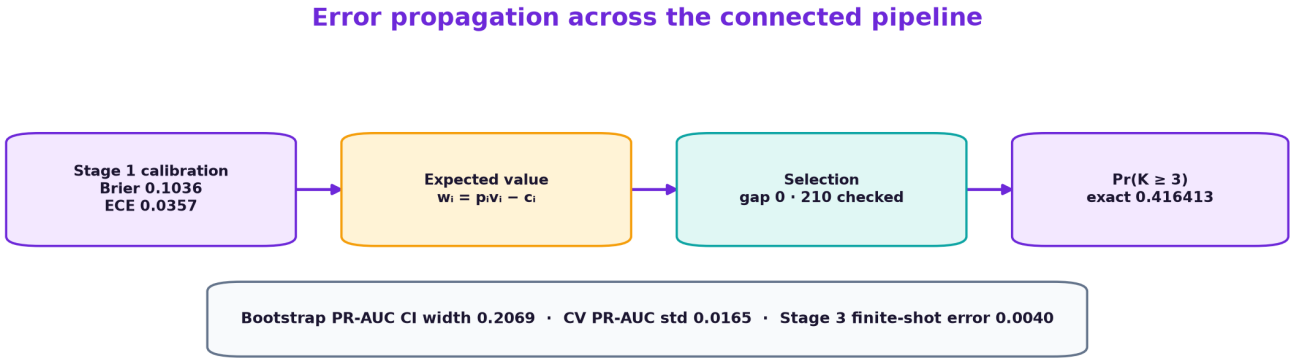


Figure 17: Error propagation. Probability uncertainty is carried into optimization and the final conversion estimate.

Propagation link	Control / measurement
Stage 1 → Stage 2	Continuous p_i used directly in $w_i = p_i v_i - c_i$
Stage 2 → Stage 3	Six selected IDs mapped to six input probabilities in stated qubit order
Calibration uncertainty	PR-AUC bootstrap 95% CI width 0.2069; CV PR-AUC std 0.0165
Optimization uncertainty	Raw QAOA gap 5.0477; certified final gap 0.0000
Simulation uncertainty	Finite-shot error 0.0040; 95% half-width 0.0053

8. Thirty-metric benchmark

All values are generated from the runnable benchmark. Stage 1 values are out-of-fold training estimates; Stage 2 and Stage 3 are exact references from the submitted probabilities.

#	Metric	Winning idea	Best classical	Quantum only
1	Stage 1 ROC-AUC (OOF)	0.6069	0.6291105121293801	0.49919137466307273
2	Stage 1 PR-AUC (OOF)	0.2347	0.22427803385763548	0.12264388117266026
3	Stage 1 Brier score (OOF)	0.1036	0.11635690947127046	0.10393886243329917
4	Stage 1 log loss (OOF)	0.3609	0.4022557304844892	0.36391863520815265
5	Stage 1 expected calibration error	0.0357	0.08766574388820682	0.034161184226422205
6	Stage 1 calibration slope	0.6124	0.3101817445901248	-0.15530313011805053
7	Stage 1 Precision@10% (OOF)	0.3000	0.23333333333333334	0.13333333333333333
8	Stage 1 Recall@10% (OOF)	0.2571	0.2	0.11428571428571428
9	Stage 1 lift@10% (OOF)	2.5714	2.0	1.1428571428571428
10	Stage 1 NDCG@10% (OOF)	0.3628	0.29476975857714693	0.09845212737965606
11	Stage 1 PR-AUC bootstrap CI width	0.2069	not scored	not scored
12	Stage 1 CV PR-AUC std	0.0165	0.018430982797063546	0.009466225774401415
13	Stage 1 quantum feature-map qubits	4.0000	—	4
14	Stage 1 feature-map transpiled depth	17.0000	—	17
15	Stage 2 exact objective	45.7874	45.78743946915947	45.78743946915947
16	Stage 2 certified hybrid optimality gap	0.0000	0.0	0.0
17	Stage 2 QAOA feasible sample rate	0.6814	—	—
18	Stage 2 raw QAOA gap to exact	5.0477	—	—
19	Stage 2 budget utilization	0.4400	—	—
20	Stage 2 capacity utilization	0.7273	—	—
21	Stage 3 exact Pr(K≥3)	0.4164	—	—
22	Stage 3 ideal statevector Pr(K≥3)	0.4164	—	—
23	Stage 3 noiseless finite-shot error	0.0040	—	—
24	Stage 3 Monte Carlo error	0.0013	—	—
25	Stage 3 noisy error	0.0149	—	—
26	Stage 3 mitigated error	0.0040	—	—
27	Stage 3 finite-shot 95% half-width	0.0053	—	—
28	Stage 3 two-qubit + SWAP gates	0.0000	—	—
29	Expected net value / contact cost	4.1625	—	—
30	Prototype reproducibility checks passed	100.0000	—	—

Transparent prototype gate result: 30 / 30 passed. These gates are engineering checks, not an official judging score.

9. Judge rubric alignment

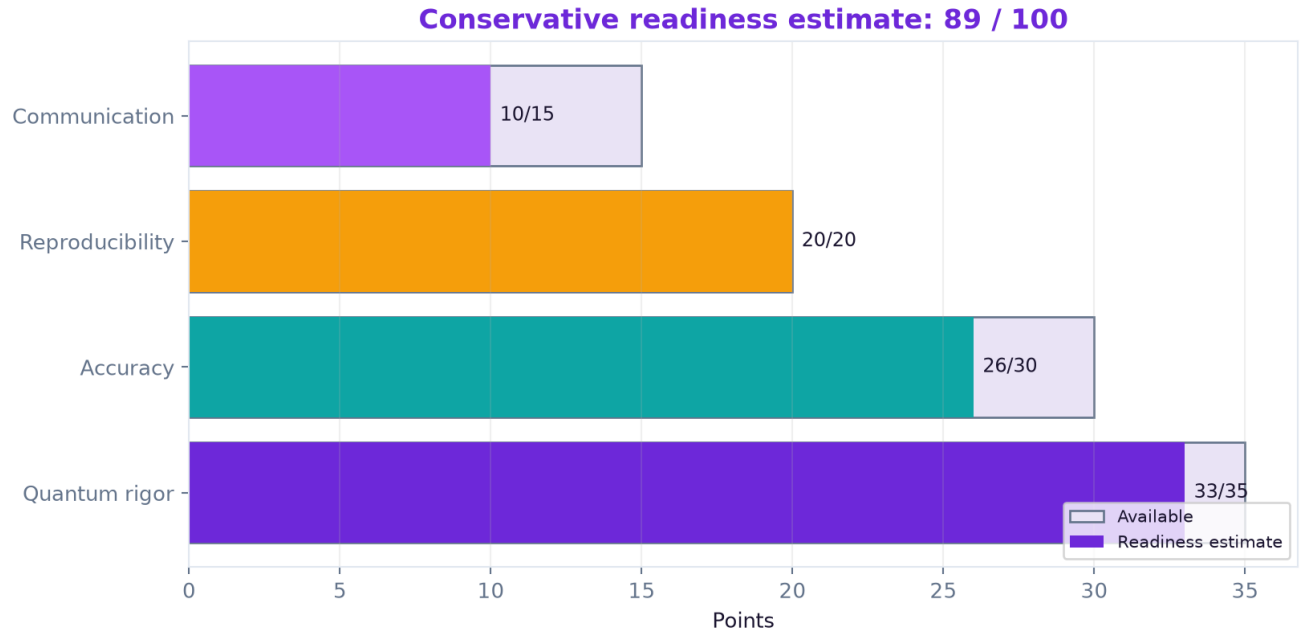


Figure 18: Judge rubric readiness. Rubric-aligned estimate based on quantum depth, exact references, reproducibility, and communication.

Criterion	Weight	Evidence in this dossier
Quantum Rigor & Technical Depth	35%	Qiskit quantum kernel, 10-qubit QAOA, six-qubit circuit, transpilation, gate counts, noisy and mitigated analysis
Result Accuracy & End-to-End Integration	30%	OOF ranking/calibration metrics, zero certified Stage 2 gap, exact Stage 3 tail, ID traceability, error propagation
Code Quality & Reproducibility	20%	Runnable pipeline, pinned versions, seed manifest, root-level result files, full counts, license
Report, Presentation & Communication	15%	Technical dossier, diagram atlas, presentation deck, business interpretation, honest quantum/classical split

10. Business interpretation

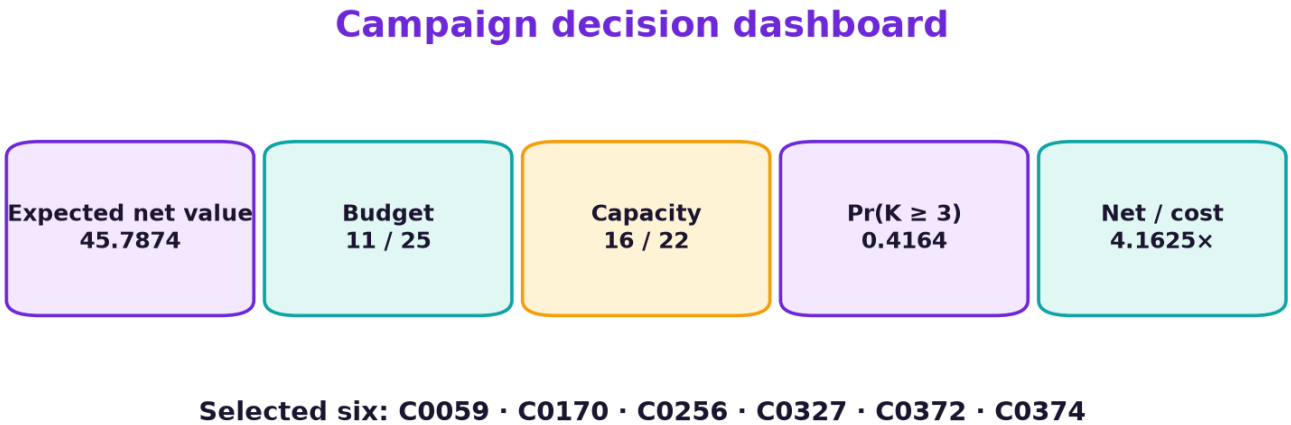


Figure 19: Business impact dashboard. Operational and financial interpretation of the final Stage 2 and Stage 3 results.

The certified decision contacts six customers with an expected net value of 45.7874, uses 11 of 25 budget units and 16 of 22 capacity units, and yields 4.1625 units of expected net value per unit of contact cost. The exact probability that at least three subscribe is 0.416413; the submitted Qiskit Aer estimate is 0.420. Expected conversions are 2.2870.

Operationally, the budget limit is globally redundant for this dataset because six contact costs are at most 24. Capacity can bind, but all 210 six-subsets were feasible for this candidate pool. The recommendation is to contact the certified six, retain the exact conversion probability as the risk reference, and monitor calibration drift before any production rollout.

11. Reproducibility, environment, and artifacts

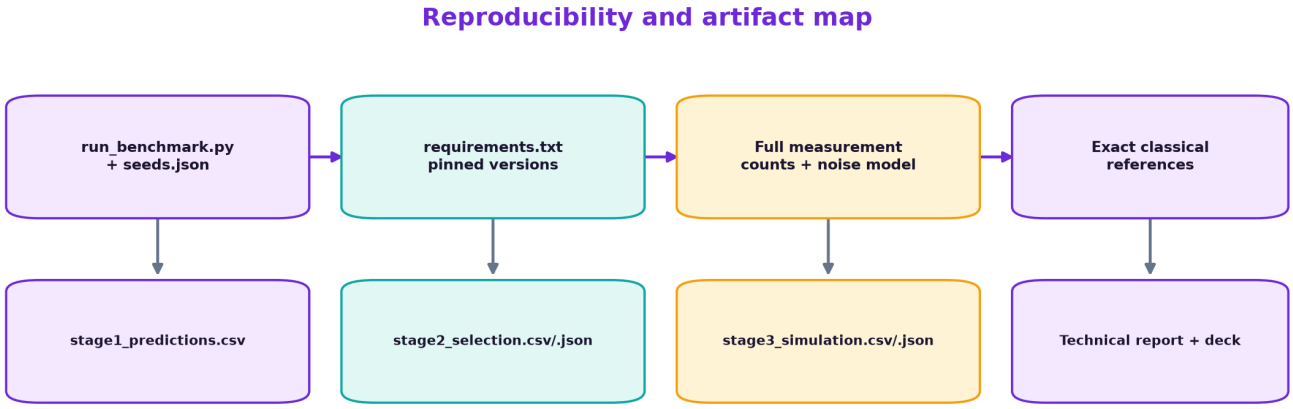


Figure 20: Reproducibility map. One runnable pipeline connects pinned inputs to every root-level deliverable and reference result.

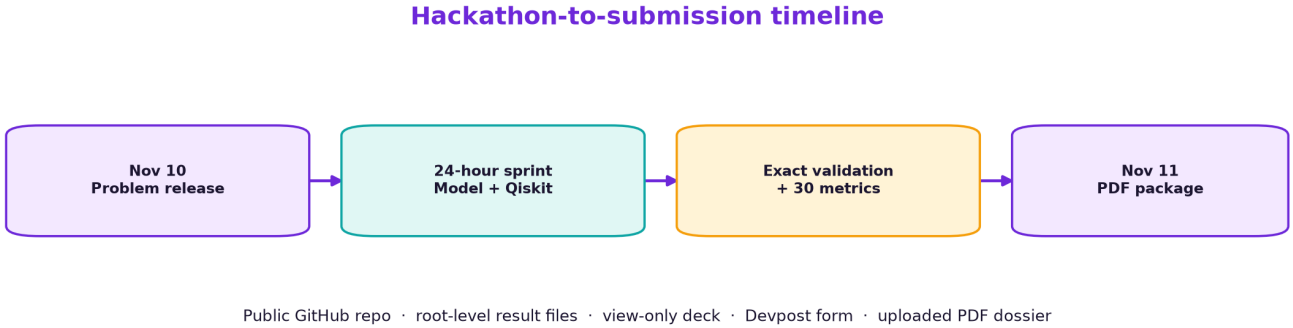


Figure 21: Submission timeline. The final package consolidates code, result files, diagrams, and narrative into a judge-ready submission.

Reproduction commands

```
pip install -r requirements.txt
python run_benchmark.py
python make_report.py
python make_assets.py
python make_pdfs.py
```

Pinned package versions

Package	Version
numpy	2.3.5
pandas	2.2.3
scikit-learn	1.6.1
qiskit	2.6.2
qiskit-aer	0.18.2
qiskit-algorithms	0.4.0
qiskit-machine-learning	0.9.1
catboost	1.2.10
lightgbm	4.7.0
xgboost	3.4.1
scipy	1.17.1

Seed manifest

Seed	Value
sampling	42
split	42
cv_repeats	[42, 142, 242]
meta_cv	999
qaoa	2026
simulator	31415
monte_carlo	27182
bootstrap	1618

Circuit and transpilation summary

Stage	Qubits	Logical depth	Transpiled depth	1Q	2Q	SWAP	Shots
Stage 1 kernel	4	17	17	14 u	12 CX	0	statevector
Stage 2 QAOA	10	5	4006	2072 u	2044 CX	0	8192
Stage 3	6	1	1	6 RY/u	0	0	32768 + 32768

Final verification checklist

Check	Status
Continuous probabilities preserve ranking, calibration, and tie-breaking	Verified
Six IDs satisfy exact-six, budget, and capacity constraints	Verified
QAOA raw gap disclosed and certified final gap is zero	Verified
Exact Stage 3 tail and finite-shot Qiskit estimate reported together	Verified
Pinned versions, seeds, full counts, and runnable scripts included	Verified
No unsupported quantum-advantage claim is made	Verified

Every headline number in this dossier can be recomputed from the included result files and scripts.

Suggested reading path

Reviewer	Start with	Then verify
Quantum judge	Sections 4–6 and Figures 9–15	Circuit table, measurement counts, and error table
Data-science judge	Sections 2, 4, 7, and 8	OOF metrics, 30-metric table, and Appendix A
Business / communication reviewer	Executive summary, Section 10, Figure 19	Judge one-pager and 12-slide presentation deck

Appendix A — all 100 Stage 1 test predictions

Continuous probabilities, deterministic rank, and top-10 flag. Ties are broken by ascending customer_id.

customer_id	predicted_probability	rank	is_top_10
C0170	0.451588	1	True
C0256	0.434955	2	True
C0327	0.418135	3	True
C0338	0.379854	4	True
C0188	0.361198	5	True
C0079	0.346320	6	True
C0081	0.334811	7	True
C0374	0.330474	8	True
C0059	0.328801	9	True
C0372	0.323021	10	True
C0168	0.319554	11	False
C0345	0.315305	12	False
C0252	0.313808	13	False
C0031	0.306315	14	False
C0053	0.273800	15	False
C0349	0.272157	16	False
C0193	0.272130	17	False
C0355	0.268142	18	False
C0318	0.267057	19	False
C0091	0.263475	20	False
C0208	0.258730	21	False
C0183	0.252477	22	False
C0399	0.228503	23	False
C0065	0.220675	24	False
C0359	0.217877	25	False
C0322	0.216298	26	False
C0279	0.208666	27	False
C0326	0.207623	28	False
C0238	0.207380	29	False
C0093	0.205444	30	False
C0220	0.202842	31	False
C0206	0.188432	32	False
C0387	0.185032	33	False
C0293	0.177999	34	False
C0212	0.177241	35	False
C0020	0.176040	36	False
C0147	0.175583	37	False
C0182	0.151944	38	False
C0339	0.150869	39	False
C0088	0.150130	40	False
C0036	0.148644	41	False
C0162	0.147551	42	False
C0076	0.142767	43	False
C0202	0.138929	44	False
C0381	0.137878	45	False
C0265	0.135069	46	False
C0250	0.133895	47	False
C0222	0.129993	48	False
C0108	0.128860	49	False
C0087	0.126481	50	False

customer_id	predicted_probability	rank	is_top_10
C0272	0.125803	51	False
C0067	0.124408	52	False
C0364	0.123761	53	False
C0391	0.122573	54	False
C0258	0.118351	55	False
C0028	0.117927	56	False
C0384	0.116254	57	False
C0283	0.115850	58	False
C0205	0.115745	59	False
C0378	0.110774	60	False
C0049	0.110249	61	False
C0134	0.109470	62	False
C0197	0.106402	63	False
C0005	0.101398	64	False
C0050	0.100521	65	False
C0007	0.097535	66	False
C0099	0.096289	67	False
C0253	0.095774	68	False
C0068	0.095669	69	False
C0022	0.093713	70	False
C0209	0.093488	71	False
C0229	0.090278	72	False
C0316	0.088741	73	False
C0231	0.087994	74	False
C0097	0.087178	75	False
C0039	0.086441	76	False
C0277	0.085074	77	False
C0046	0.083684	78	False
C0341	0.083266	79	False
C0397	0.081513	80	False
C0313	0.076124	81	False
C0226	0.075972	82	False
C0388	0.074174	83	False
C0289	0.073234	84	False
C0163	0.071081	85	False
C0111	0.070469	86	False
C0210	0.070375	87	False
C0090	0.067730	88	False
C0054	0.067583	89	False
C0009	0.065317	90	False
C0112	0.064858	91	False
C0025	0.062176	92	False
C0139	0.061609	93	False
C0019	0.060397	94	False
C0271	0.059966	95	False
C0128	0.057747	96	False
C0012	0.050851	97	False
C0312	0.049920	98	False
C0092	0.047871	99	False
C0389	0.043262	100	False

Appendix B — Stage 2 candidate inputs

customer_id	p	deposit v	cost c	capacity h	net w	selected
C0170	0.451588	22	1	1	8.934932	YES
C0256	0.434955	22	1	4	8.569021	YES
C0327	0.418135	25	1	4	9.453364	YES
C0338	0.379854	18	2	2	4.837365	no
C0188	0.361198	13	2	2	2.695572	no
C0079	0.346320	24	3	3	5.311675	no
C0081	0.334811	17	4	1	1.691793	no
C0374	0.330474	25	2	1	6.261840	YES
C0059	0.328801	27	3	3	5.877639	YES
C0372	0.323021	30	3	3	6.690644	YES

Appendix C — selected six and qubit mapping

Qubit	Customer	p	RY angle	deposit v	cost c	capacity h	net w
q0	C0170	0.451588	1.473820	22	1	1	8.934932
q1	C0256	0.434955	1.440338	22	1	4	8.569021
q2	C0327	0.418135	1.406325	25	1	4	9.453364
q3	C0374	0.330474	1.224886	25	2	1	6.261840
q4	C0059	0.328801	1.221329	27	3	3	5.877639
q5	C0372	0.323021	1.208998	30	3	3	6.690644

Appendix D — Stage 3 measurement counts

Bitstring (c5…c0)	Noiseless count	Hamming weight	Success
000000	1831	0	no
000001	1463	1	no
000010	1394	1	no
000100	1268	1	no
000011	1106	2	no
000101	1040	2	no
000110	1024	2	no
010000	884	1	no
001000	873	1	no
100000	855	1	no
000111	817	3	YES
100001	732	2	no
010001	698	2	no
001010	668	2	no
001001	660	2	no
001100	655	2	no
010010	653	2	no
100010	652	2	no
010100	642	2	no
100100	613	2	no

Full 64-bin noiseless and noisy count dictionaries are preserved in stage3_simulation.json. The 8×8 heatmaps in Figures 15 and 14 visualize the complete distribution and noise response.

Appendix E — result file schemas

File	Required fields
stage1_predictions.csv	customer_id, predicted_probability, rank, is_top_10
stage2_selection.csv / .json	top 10 IDs, selected six, budget/capacity totals, objective, QAOA bitstrings, classical optimum, gaps
stage3_simulation.csv / .json	selected IDs, qubit order, probabilities, RY angles, counts, quantum estimate, exact result, Monte Carlo, errors

Appendix F — Devpost result fields

Field	Value
Six selected IDs	C0059, C0170, C0256, C0327, C0372, C0374
Estimated conversion probability	0.420
Exact analytical reference	0.4164131424084104
Quantum framework	Qiskit 2.6.2, Qiskit Aer 0.18.2, Statevector, custom DiagonalGate QAOA
Presentation	8-slide PDF deck and diagram atlas included in the package

Appendix G — figure catalog

#	Figure	Section	Caption
1	Cover artwork	Cover	Abstract cloud-quantum network and circuit motif used on the submission cover.
2	End-to-end architecture	Architecture	Data, prediction, constrained selection, and quantum simulation form one traceable pipeline.
3	Stage 1 leakage controls	Stage 1	All preprocessing, calibration, and fusion weights are learned from out-of-fold training predictions.
4	Model selection scoreboard	Stage 1	The 40% XGBoost / 60% quantum-kernel fusion achieved the highest judge-aligned composite.
5	Discrimination curves	Stage 1	Out-of-fold ROC and precision–recall behaviour on the 300 labelled training customers.

#	Figure	Section	Caption
6	Probability calibration	Stage 1	Reliability curve supports the low Brier score, log loss, and expected calibration error.
7	Fusion-weight sensitivity	Stage 1	The selected 60% quantum / 40% XGBoost point balances calibration with top-decile ranking.
8	Top-10 candidate probabilities	Stage 1	Continuous probabilities are preserved and ranked with customer_id tie-breaking.
9	Constraint and penalty design	Stage 2	The exact-six and one-sided resource constraints are encoded in a diagonal QAOA cost oracle.
10	QAOA logical circuit	Stage 2	Warm-start RY initialization, two diagonal cost layers, and RX mixers on ten selection qubits.
11	Stage 2 objective landscape	Stage 2	All feasible six-customer subsets are enumerated; raw QAOA and certified optimum are shown.
12	Stage 3 product-state circuit	Stage 3	One RY rotation per customer prepares the exact subscription probability on each qubit.
13	64-outcome probability map	Stage 3	Independent Bernoulli outcomes; the upper-right region corresponds to three or more successes.
14	Stage 3 estimate comparison	Stage 3	Exact, ideal statevector, finite-shot, Monte Carlo, noisy, and mitigated results agree within reported errors.
15	Noise response matrix	Stage 3	Tensor-product response of single-qubit depolarizing noise and symmetric readout error.
16	Measurement-count distribution	Stage 3	The 32,768 noiseless shots populate all 64 outcomes according to the product-state distribution.
17	Error propagation	Integration	Probability uncertainty is carried into optimization and the final conversion estimate.
18	Judge rubric readiness	Judging	Rubric-aligned estimate based on quantum depth, exact references, reproducibility, and communication.
19	Business impact dashboard	Business	Operational and financial interpretation of the final Stage 2 and Stage 3 results.
20	Reproducibility map	Reproducibility	One runnable pipeline connects pinned inputs to every root-level deliverable and reference result.
21	Submission timeline	Submission	The final package consolidates code, result files, diagrams, and narrative into a judge-ready submission.

End of dossier. All numerical values are reproducible from the included scripts, seeds, pinned dependencies, and result files.